PARTH SETHI

Senior Software Engineer

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SUMMARY

Senior Software Engineer with 6+ years in system design and high-throughput, fault-tolerant distributed systems and gRPC microservices. Built the core engine behind a $1B+ revenue channel, drove a 10x throughput gain, and raised availability to 99.999% across AWS and Salesforce.

PROFESSIONAL EXPERIENCE

Salesforce **May 2022 - Present**

Senior Software Engineer (SMTS), Seattle, WA

- **Architected "Sage" from scratch**, a two-step gRPC microservices rendering engine replacing the legacy E360 platform - now the core engine behind a **$1B+** revenue channel; shipped critical functionality two releases ahead of schedule.
- Set a reference architecture adopted by **multiple teams** - Sage is now the shared engine parsing scripts across SMS, email, and RCS channels.
- **Built an agentic automation pipeline (Claude)** that replaced a manual **~4-hour-per-org** workflow - org provisioning, preview/real email sends, and integration + daily health tests - reclaiming multiple engineer-weeks per quarter.
- Drove a **10x throughput gain** (10M to 40M emails/hour) through DynamoDB batch tuning and Redis caching and deduplication.
- **Built a pre-deploy validation framework** that caught regressions before rollout, improving fault tolerance and raising availability from 98% to **99.999%**; on call, cut incident response load and PagerDuty alerts by 10+/week.
- Drove **100% functional parity** with the legacy platform via a custom Compatibility Testing Framework, in cross-functional collaboration with Product, to de-risk the E360-to-Sage migration.
- Rebuilt email processing into **bulk operations** with real-time, quota-aware entitlements, enabling partial sends under load.

Amazon Web Services (AWS) **Oct 2019 - May 2022**

Software Development Engineer, Seattle, WA

- **Rearchitected the EC2 maintenance service** to fire "reboot" instance events across 25 public regions, protecting **556,000+** customers' instance data on degraded hardware.
- **Redesigned EC2 maintenance scheduling** with a best-fit algorithm, eliminating daily manual scripts and cutting "no time slot" errors by 100s every two weeks.
- **Automated manual maintenance escalations** on internal EC2 fleets, removing **4,000 escalations/year** and hundreds of engineer-hours.
- **Designed a customer-window scheduling algorithm** giving 10,000+ customers control over maintenance timing, plus an EC2 KPI "pain index" to predict and reduce escalations.

OM Partners **Feb 2019 - Oct 2019**

Software Engineer, Atlanta, GA

- **Designed a task-distribution scheduling algorithm** over time buckets from user input, optimizing work-execution rate.
- Implemented optimization strategies in **C++17** for "OPAL," a custom supply-chain inventory and logistics language.

TECHNICAL SKILLS

Languages: Java, C++, SQL, NoSQL (proficient); C#, Scala, Python, Ruby (familiar)

AWS: DynamoDB, Redshift, RDS, S3, Lambda, Kinesis, QuickSight, EC2, CloudWatch, CloudFormation, IAM, KMS, SNS, SQS

Systems & Tools: gRPC, Redis, distributed systems, high-availability design, performance optimization, Git, Agile/Kanban, AI-agent orchestration (Claude, Cursor)

EDUCATION

M.S. Computer Science, University of Florida, Gainesville, FL Aug 2017 - Dec 2018

GPA 3.56 / 4.0

B.Tech Electronics & Communication, Jaypee University of Info. & Tech., Solan, India Aug 2013 - May 2017

GPA 9.1 / 10